SOLUTION

1. Chloroform mixed with nitrogen gas is an example of

(1) Gas in Gas solution

(2) Liquid in Gas solution

(3) Solid in Gas solution

(4)Liquid in Liquid solution

2. Concentration of pollutants in water or atmosphere is often expressed in terms of

(a) Molarity (b) Molality (c) $\mu g/ml$ (d) ppm

(1) a & b (2) a & d (3) b & c (4) c & d

Note – Mass %, ppm, mole fraction and molality are independent of temperature.

3. Scuba divers cope up with the situation of bends by breathing in

(1) Pure air

(2) Pure oxygen

(3) Air diluted with helium

(4) Air diluted with Oxygen

4. Raoults law can be said to be a special case of Henry's law in which K_H (Henry's Constant) becomes equal to –

(1) Mole fraction of solute

(2) Mole fraction of solvent

(3) vapour pressure of volatile component

(4) Total vapour pressure

5. Molecular mass of polymers is determined by using

(1) Relative lowering in vapour pressure

(2) Elevation in boiling point

(3) Depression in freezing point

(4) Osmotic pressure

6. Normal saline solution which is isotonic with blood cells

(1) 1N salt solution

(2) 0.9% mass/volume NaCl solution

(3) 1N NaCl solution

(4) Normal salt solution